QUIVER 1.6 · TECHNICAL WHITE PAPER

True Effort Score

A transparent, multi-signal model of running load on Apple Watch

Wayne Bishop
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SUMMARY

True Effort Score (TES) measures athletic effort from six signals derived from Apple Watch. These include heart rate, pace, cadence, grade, vertical oscillation and altitude. A baseline model learns the heart rate of a runner's anticipated workload and reports any potential gap. A classifier model also labels each moment based on how the runner is moving. As a result, efforts that require significant biomechanical effort (e.g., descending downhill) are designed to count as hard work. Designed for watchOS, TES is built in Swift as a model within the open-source Quiver framework.

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1 History

In the 1970s researchers at Simon Fraser University in Burnaby, British Columbia, Canada, proposed a model of how training performance calculations should work. The idea was based on **training** being the input and **performance** the output. Each workout is reported as an "impulse" that produces two responses: fitness, which builds slowly and lasts, and fatigue, which builds quickly but eventually fades. Performance on any day is roughly fitness minus fatigue. This became known as the fitness-fatigue model, or the impulse-response model. To manage it they defined the term **training impulse** or TRIMP to determine the significance of each workout. The first version scored a swimmer's training in points for every 100 meters, weighted by intensity [1]. By 1990, the number came from heart rate: the minutes of a workout, weighted by cardiovascular intensity [7].

Over the years the concept was refined. In 1993, a simpler scheme called **summated heart-rate zones** appeared in coaching literature. The possible range of heart rate was divided into five zones, and the minutes spent in each zone were multiplied by the zone's number from one to five [13].

Introduced in 2001, a method named **session-RPE** determined training load without relying on heart rate data. Building on a perceived exertion scale from the 1960s, later revised to ten points [14], the calculation involved multiplying the athlete's self-reported exertion by the session's duration in minutes [4]. What this approach sacrifices in biometric precision, it gains in universal reach. It applies to any sport, requires no hardware sensors, and captures systemic fatigue that heart rate alone cannot measure.

1.1 The running boom

Fast forward to today and most fitness watches calculate training load from one or a combination of these ideas. Heart-rate zones remain the default on fitness watches, while perceived effort can be seen in the post-workout rating many apps now request. In many ways both approaches utilize the TRIMP's formula of intensity multiplied by time.

In many respects, TRIMP provides a good approach to measuring cardiovascular effort. However, the sport of running has grown in both depth and participation since the 1970s. Today, even though less than 1% of people have ever run a marathon, 2027 will mark the first occasion where a world marathon major (London) spans two days, welcoming 100,000 participants across the weekend [15]. For the first time in history, we are seeing runners complete the 26.2-mile distance in under 2 hours [16]. And in the trail running world, the fastest runners at the Ultra-Trail du Mont Blanc now complete the roughly 108-mile course with nearly 32,000 feet of elevation gain in about 19 hours [17]. Runners at every level now train across a far wider range of terrain, intensity and recovery than the steady efforts TRIMP was built to measure. A score based on heart rate and time alone was never designed for a steep mountain descent, a set of hill repeats or a careful return from injury.

2 Understanding true effort

While advancements in both nutrition and shoe technology have certainly contributed to running's success, arguably the biggest advancement has come from training principles. As runners know, successful training requires balancing effort and recovery. Knowing how hard to push on any given day dictates the difference between success and injury. Today, runners develop not only cardiovascular excellence but also muscular endurance, extending to their tendons and ligaments. Examples include:

- **Running downhill.** On a steep descent the quadriceps contract eccentrically to brake every foot strike. Compared to running on a flat surface, a –9° (about 16%) slope produces roughly 50% more impact force and over 70% more braking force with each stride [9], and the muscles, ligaments and tendons absorb that added load. Because gravity supplies much of the work, the effort may not register as significant cardiovascular activity [10].
- **Interval training.** Short periods of intense effort are separated by walking, standing, or easy recovery jogging. While surges demand significant cardiovascular output, averaging them together with the rest periods masks the true intensity.
- **Technical terrain.** Rocks, roots and loose footing demand concentration, agility and flexibility on every stride, whether climbing, descending or on the flat. Pace slows and heart rate may stay moderate, so neither reflects the added work.
- **Power-hike.** On a steep climb a runner may drop to a hike. Pace and cadence collapse while cardiovascular effort increases. Scores solely based on pace undercount this effort.
- **Environmental factors.** In heat, at altitude, or late in a long run, heart rate climbs while the external workload holds steady. This is known as cardiac drift, or decoupling [6]. A heart-rate score may read higher than the workload warrants.
- **Returning from injury.** Injuries often involve a stress fracture or damage to a muscle, ligament or tendon. Measuring progress isn't about increasing cardiovascular effort but about carefully measuring biomechanical load, usually through a return-to-run or walk-run program.

3 Running power

The other established approach is running power, now offered on most sports watches and measured more directly by foot pods. A power model reduces a run's biomechanics to a single number in watts and derives training load from it in the tradition of power-based stress scores [2]. That load is measured against the runner's critical power: the highest power they can sustain without tiring, roughly what they could hold for 30 to 60 minutes. In simplified form, power scales with the runner's mass and speed, plus the work of climbing or descending:

$$\textbf{power} \approx \text{mass} \times \text{speed} \times (\text{cost of running} + g \times \text{grade})$$

Here the cost of running is the energy spent per kilogram for each meter on flat ground [10], *g* is gravity, and grade is the slope, which turns negative on a descent.

On a steep descent, estimated power drops: the runner produces fewer watts even as the quadriceps absorb heavy eccentric load, so they may receive an overall low score. On a hot day, power never sees the heart, so the environmental cost goes unrecorded.

The comparison cuts both ways. A foot pod records better raw biomechanics than any wrist sensor, and for steady-state running a power-based load has years of field use behind it. Because a foot pod measures each stride with its own motion sensors, its numbers depend on neither GPS nor heart rate. A runner gets consistent readings outdoors, on a level treadmill, or where the GPS signal is poor. In testing, the best-performing foot pod was the most repeatable of five running-power technologies both indoors and outdoors [11].

4 Introducing True Effort Score

True Effort Score (TES) captures each moment of a run as a vector of six signals. These include heart rate, pace, cadence, grade, vertical oscillation and altitude. TES keeps these signals together rather than reducing them to a single number at the start. TES ships as a machine learning model in Quiver, the open-source Swift ML framework.

A vector is a set of numbers treated as one unit. Each number represents one aspect of the moment, and together they position that moment in space. The distance between two vectors then shows their similarity: moments that look alike sit close together, moments that differ sit far apart.

A single number can make distinct efforts appear identical. A strong effort on flat ground and a rapid descent can share nearly the same pace, while their heart rate and vertical oscillation pull in opposite directions. As vectors, the two moments remain distinct:

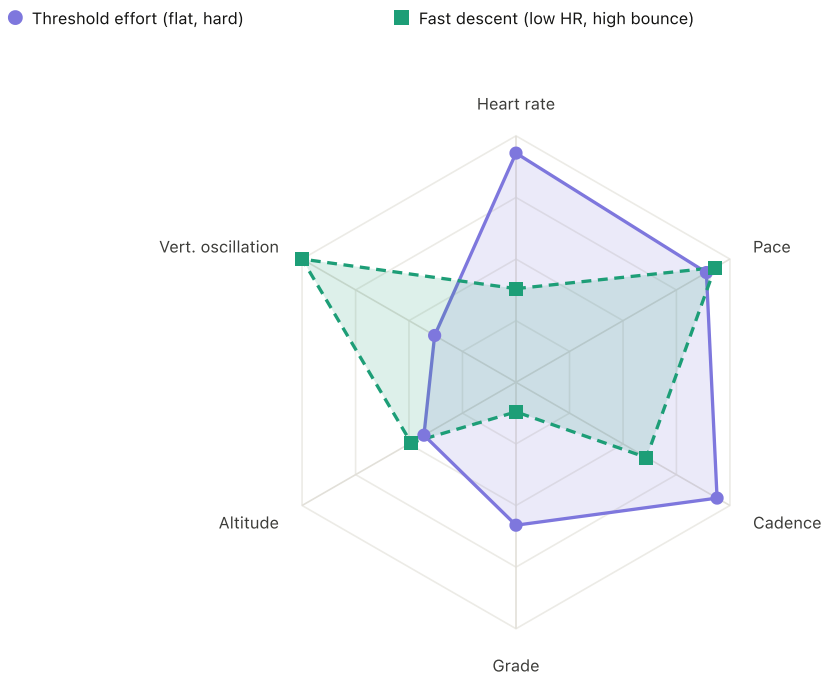


Figure 1. Two moments of a run as six-signal vectors: a threshold effort on flat ground and a fast descent. Values are illustrative and scaled to a common range.

TES also uses distance to classify effort. Its classifier, a nearest-neighbor model, labels each moment by finding the labeled examples closest to it. For more on vectors, see the Linear Algebra Primer in the Quiver documentation.

4.1 How it works

TES asks two questions at each moment. First, what should the heart be doing for this workload? A personal baseline, learned from the runner's own history, predicts the heart rate the workload should produce. The gap between this prediction and the actual reading is **the residual**: heart rate the workload doesn't explain (see The residual).

Second, what kind of effort does each moment look like? A classifier compares the movement pattern against known efforts and labels it **Recovery**, **Easy**, **Tempo**, **Threshold**, or **Hard** (see Effort categories). The score adds up these labeled moments, weighted by intensity and duration, plus a few costs that only exist across a whole run (see The score).

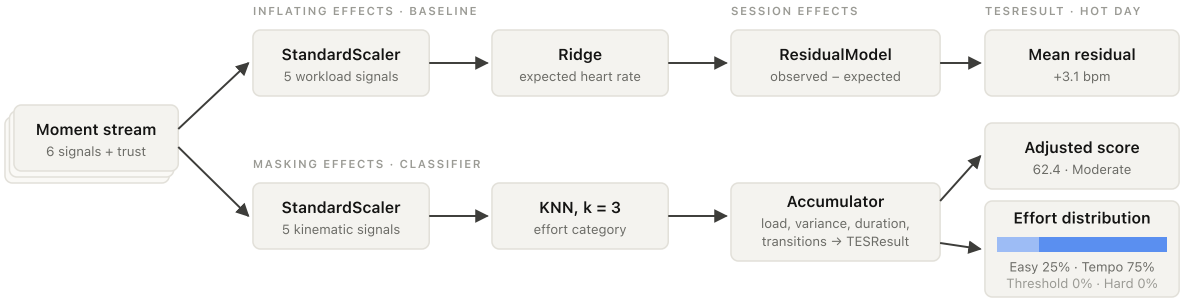


Figure 2. The two branches of TES: the baseline and the classifier.

Optical heart-rate sensors are prone to interference from a loose fit, cold skin, or cadence lock—a phenomenon where the sensor locks onto the runner's foot strikes rather than their pulse [8]. TES mitigates this by applying a confidence weight to every sample, ensuring that degraded readings have less influence on the final score.

4.2 Scaling signals

The data arrive in very different units. Heart rate and cadence values often sit in the 150s or 170s, altitude can run into the hundreds of meters, and grade moves by only a few percentage points. If left in raw units, the largest numbers would dominate. The classifier's distances would be decided almost entirely by heart rate and cadence, and a meaningful change in grade would barely register.

Before either branch reads a moment, we use a **StandardScaler** to subtract each signal's average and divide by its standard deviation—the typical variation around that average:

$$\textbf{scaled value} = \frac{\text{value} - \text{average}}{\text{standard deviation}}$$

In this scaled space, zero represents the typical value, +1 is one standard deviation above the average, and –1 is one below. For example, if a runner's cadence averages 170 steps per minute with a typical variation of 4, and their grade averages 1% with a typical variation of 2%, a cadence of

176 and a grade of 4% both scale to +1.5. The original units differ, but both scaled values sit equally far from typical.

Each branch scales its own set of five signals. The baseline leaves out heart rate (the value it predicts) and learns its averages from the runner's last four weeks of history, so "usual" is personal. The classifier leaves out altitude and learns its scale from the known efforts it compares against. Scaling is also what makes the baseline's coefficients comparable (see The baseline).

4.3 What it solves

Traditional systems rely on a single signal, either heart rate or power estimated from speed, grade and mass, to interpret an entire run. TES works the other way around. It reads all six signals across its two branches, labels each moment from the classifier's five, then sums the labeled moments into a score.

This lets the scenarios described earlier function as intended. Descents count because the classifier evaluates a full movement pattern rather than one potentially misleading signal, and power-hikes count because the walking gate grades them by climb rate and heart rate. Intervals count at full intensity because each moment is labeled individually, so a running surge reads as a surge regardless of the recovery segment. Finally, heat and cardiac drift are separated into a residual beside the score. After each outdoor run, the baseline refits from the runner's recent history, so the baseline's expectations keep pace with the runner's fitness.

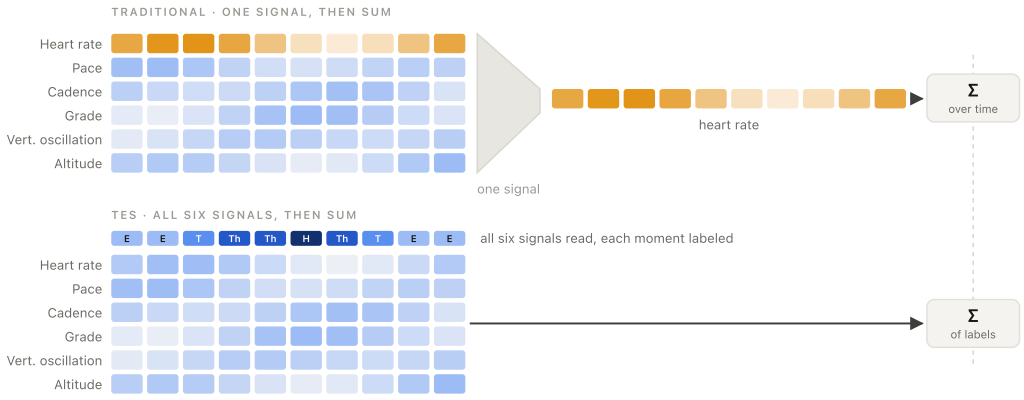


Figure 3. Traditional scores pick one signal — heart rate, or power derived from speed, grade and mass — then sum it over time. TES reads all six signals across two branches, labels each moment, then sums. Values are illustrative.

4.4 The baseline

The **baseline** learns the heart rate a runner's workload should produce, from the last four weeks of that runner's history. It is a ridge regression [5], a linear regression with a light penalty that keeps the learning stable when signals overlap, learned from **pace**, **cadence**, **grade**, **vertical oscillation**

and **altitude**. Unlike other systems the calculation doesn't require knowledge of one's age, heart-rate zones or other demographic inputs. The baseline is developed from the runner's own history. For example, from a sample 12-run history:

$$\begin{aligned}\textbf{expected HR} = 142.69 &- 4.624\,\text{pace} + 2.871\,\text{cadence} + 4.853\,\text{grade} \\ &- 3.952\,\text{vertical oscillation} + 0.5405\,\text{altitude}\end{aligned}$$

The first number, 142.69, is this runner's average heart rate across their history: the heart rate expected when every signal sits at its usual value. Each **coefficient** then adds or subtracts beats per minute as its signal moves one standard deviation, a typical swing for this runner, above its average. Because every signal is put on the same scale (see Scaling Signals), their sizes can be compared directly. For this runner, grade counts about **nine times more** than altitude. Pace is negative because it is measured in minutes per kilometer, so a larger number means slower running and a lower heart rate. Vertical oscillation is also negative: for this runner, a bouncier stride came with easier running, while the flattest strides came on hard climbs.

4.5 The residual

By comparing the expected heart rate against measured heart rate, we can see the effects of other factors. A moment one standard deviation steeper than usual, with every other signal at its average, is expected to read 142.69 + 4.853 ≈ 147.5 bpm. If a watch reads 155 bpm at that time interval, the gap of +7.5 bpm is the **residual**: heart rate the workload doesn't explain. TES reports it at every interval and as an average across the run.

TES keeps the residual separate from the final score to isolate mechanical effort from environmental stress. If we combined them, the score would blur the line between a heavy physical workload and external factors like heat, fatigue, or caffeine.

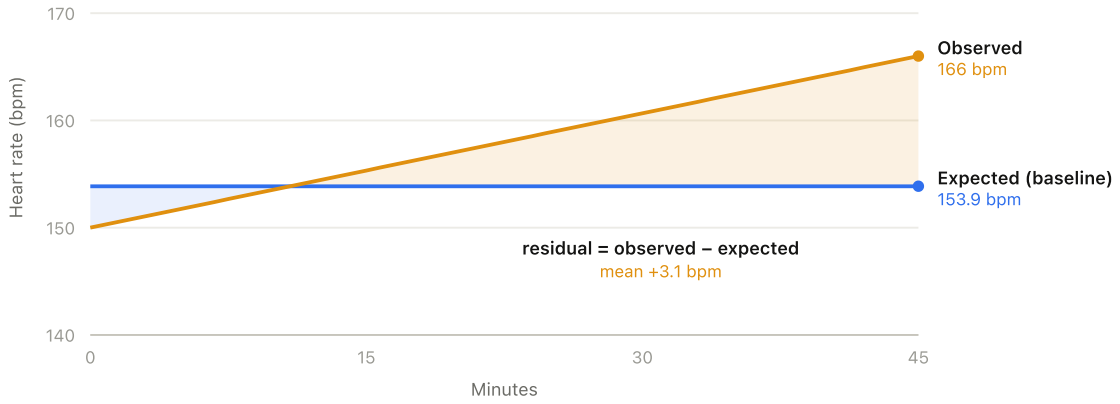


Figure 4. The hot-day scoring example: observed heart rate drifts above the baseline's expectation over 45 minutes. The shaded gap is the residual, mean +3.1 bpm.

Typical residuals shrink over time as the baseline learns a runner's profile. Because a first run has no history to compare against, TES reports no residual; the baseline is only generated after that run completes. During the next few sessions, predictions remain imprecise and residuals can be large. As history accumulates, the baseline becomes more accurate, causing typical residuals to settle near zero.

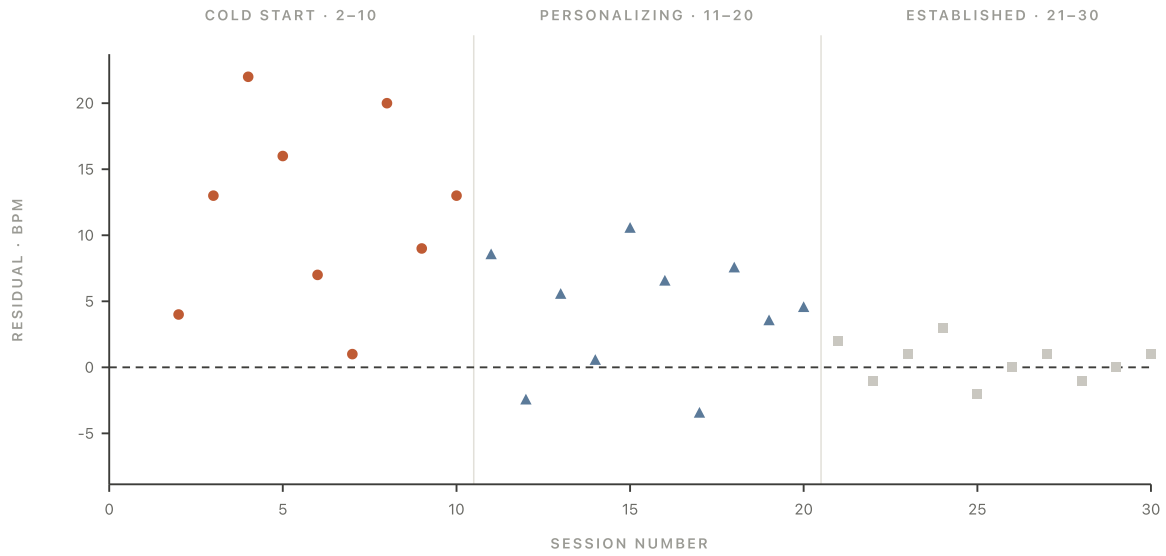


Figure 5. Session residuals shrink as the baseline learns the runner: large on early runs, smaller while personalizing, and near zero once established. Values are illustrative.

4.6 Effort categories

The classifier is a Nearest-Neighbor model [3] that compares each moment with a set of known efforts and gives it the label of the closest ones. Overall, there are five categories:

Category	Meaning
Recovery	Walking on level or gentle ground
Easy	Aerobic base
Tempo	Sustained sub-threshold, roughly marathon to half-marathon pace
Threshold	At lactate threshold
Hard	A heavy biomechanical load not reflected by heart rate **or** a sustained effort above threshold

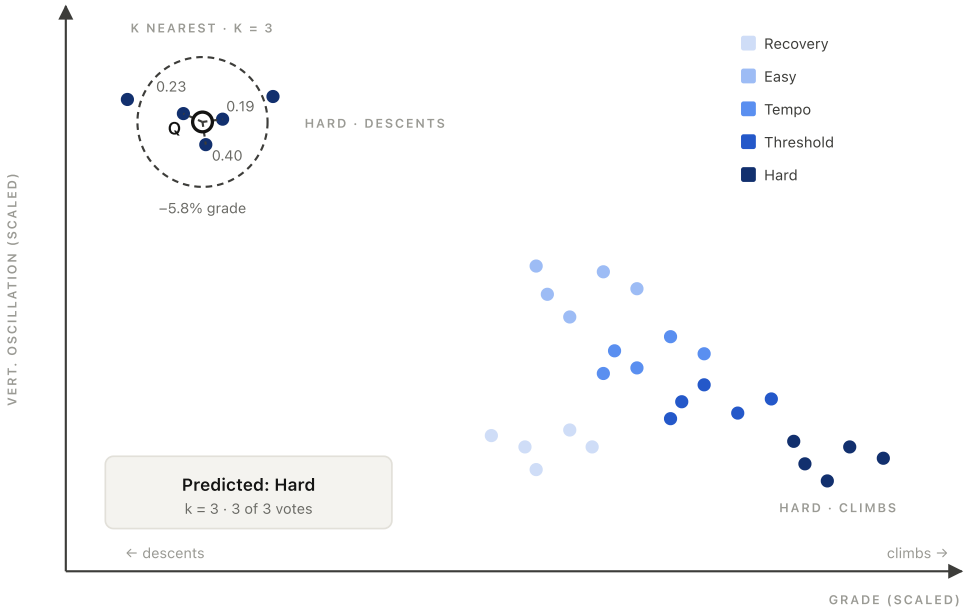


Figure 6. A new moment Q is labeled by majority vote of its three nearest known efforts: all three are Hard descents, so Q reads Hard. Shown on grade and vertical oscillation, two of the five scaled classifier signals; the model votes in the full five-signal space. Hard appears as two clusters (descents and climbs) because the category names load, not a heart-rate zone. Values are illustrative.

A time interval on a steep descent at 131 bpm, 5.1 min/km and −5.8% grade — moment Q above — reads **Easy** by heart rate alone. Its grade and vertical oscillation match the descent efforts in the known set, so TES labels it **Hard**.

The top category is intentionally called **Hard** rather than a cardiovascular zone name such as VO_2 max. For example, a steep eccentric descent is classified as **Hard**. This could also include a steep power-hike. Naming the category after a cardiovascular quantity would hide the biomechanical load TES is designed to capture.

4.7 The score

The raw score adds up the time spent in each category, weighting harder efforts more heavily. The scale is anchored so that one hour held at **Threshold** reads 100, following the convention of power-based training stress scores [2].

The category of **Threshold** is a repeatable effort a runner can hold for about an hour. Threshold efforts are fast running on flat to gently sloping ground. The score rises with both intensity and time and has no ceiling.

Some costs exist only across a run. **Intervals** cost more than steady running of the same average effort, and jumps between categories cost more than gradual changes, so both raise the score. Runs past 45 minutes also earn a small credit for time on feet. The credit multiplies the score rather than adding a fixed amount, so a harder run gains more from the same minutes. With that credit, the same hour at threshold reads 102.9, the headline adjusted score. Each adjustment is also reported on its own, so a reader can see what moved the score away from the raw total. In full, the adjusted score is:

$$\textbf{adjusted score} = \text{raw} \times \text{variance} \times \text{time-on-feet credit} + 0.1 \times \text{transitions}$$

Variance tracks how widely a run's efforts fluctuate across categories after an initial five-minute warm-up, increasing the score by a maximum of 35 percent. The time-on-feet credit takes effect at 45 minutes. Transitions penalize only abrupt spikes of two or more categories, such as jumping from **Easy** straight to **Threshold**, leaving gradual step-ups unpenalized.

The time-on-feet credit grows slowly on purpose, and it flattens as runs get longer. Many runners stop a long run near two hours, and for runners covering 40 or more miles a week, running coach **Jack Daniels** recommends the lesser of 150 minutes or 25 percent of weekly mileage [12]. At two hours the credit is about 10 percent; at 150 minutes, 12 percent. TES sets no maximum value: a three-hour run still earns more, just less for each added hour:

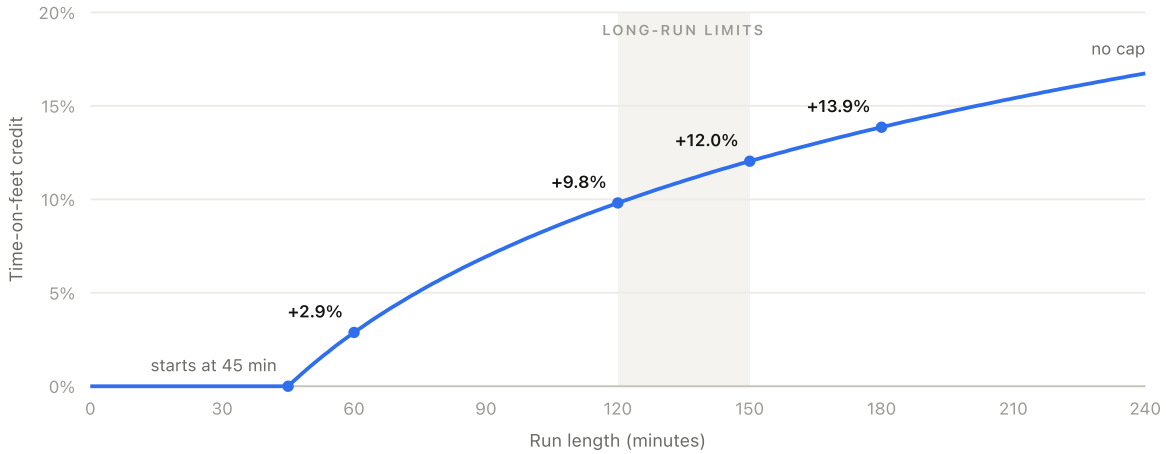


Figure 7. The time-on-feet credit starts at 45 minutes and flattens as runs get longer. The shaded band marks the usual long-run limits of two hours to 150 minutes.

Scores for easy running are intentionally low. An hour at **Easy** counts for only a third of an hour at **Threshold**. Each step up adds another third of an hour at **Threshold**, so the scale climbs evenly, and an hour at **Hard** counts for a third more than an hour at **Threshold**. TES captures the hidden costs of descents and surges by labeling those moments **Hard**, not by stretching the scale. Long runs still accumulate points over time, and the time spent in each category is reported beside the score.

Heart rate does not directly dictate the final score; rather, it functions as one of several features used to classify the effort category. To prevent environmental stress from inflating the score, once a baseline exists, the classifier caps the observed heart rate at the baseline's expected value for that workload. Readings below the expectation pass through unaltered, while low-confidence readings are mathematically regressed toward the baseline. This ensures that factors like heat or cardiac drift cannot artificially elevate a moment into a harder category.

4.8 What a score means

The adjusted score operates on an absolute scale divided into five ranges. Each boundary represents slightly more than one continuous hour in a specific effort category. Factoring in the time-on-feet credit, a steady hour yields a score of 34 at **Easy**, 69 at **Tempo**, 103 at **Threshold**, and 137 at **Hard**. As a rule of thumb, **every 35 points** represents one hour spent at the next highest

intensity.

Range	Score	Equivalent Continuous Effort
Light	< 35	Up to one hour at Easy
Moderate	35–70	Up to one hour at Tempo
Substantial	70–105	Up to one hour at Threshold
Heavy	105–140	Up to one hour at Hard
Extreme	≥ 140	More than one hour at Hard

Table 1. Named ranges on the adjusted score.

Because surges and abrupt changes in category compound the score, mixed-intensity sessions often reach higher ranges in less time than steady efforts. These labels estimate training load; they are not recovery suggestions for the next day.

A recreational runner logging 45–70 minute easy runs will generally score between 25 and 40, straddling the Light and Moderate ranges. In contrast, a mountain runner sustaining high effort on climbs and descents will routinely land in the Heavy range, such as the 117.9 downhill example seen below.

4.9 Scoring examples

Here are four hypothetical efforts that demonstrate how TES behaves in practice. Each example indicates the model state: *first run* (no personal history) or *established runner* (a learned baseline). These examples illustrate intended model behavior rather than physiological accuracy. The downhill and hot-day runs use a 5-second sampling interval, matching Apple's latest reported heart-rate frequency during workouts [8].

A downhill trail run (first run): 117.9, Heavy. A 55-minute run featuring four extended –6% descent blocks. While the runner's heart rate remains calm, movement data, matched against TES's built-in known efforts, categorizes 58 percent of the run as **Hard**. No personal history is needed: the classifier labels descents from the first run. A traditional heart-rate model would log this as an easy day; TES prices the eccentric load.

Hill repeats versus a steady Tempo (first run): 76.4 versus 52.4. Both 47-minute sessions start from nearly the same raw score. The difference comes entirely from the session adjustments, which TES reports beside the score:

Session	Raw	Variance	Time on feet	Transitions	Adjusted
Hill repeats	52.8	× 1.350	× 1.004	+ 4.8	76.4
Steady Tempo	52.2	× 1.000	× 1.004	+ 0.0	52.4

Table 2. The session adjustments for two runs with nearly the same raw score.

The eight surges raise variance to its 35 percent maximum, and every jump between **Easy** and **Hard** adds to the transition total. The steady run earns neither.

A hot day versus a cool day (established runner): 62.4 versus 62.3. Both 60-minute runs hold 45 minutes at the same steady pace between a 10-minute warm-up and a 5-minute cool-down, so the baseline expects the same heart rate for each, 148.0 bpm. On the hot day, cardiac drift drives the heart rate from 150 to 166 bpm over those 45 minutes. Because the work is the same, TES does not let the extra heart rate raise the effort category, and the score barely changes. Instead, the extra beats appear in the residual, the gap between actual and expected heart rate: +3.1 bpm on the hot day versus –2.8 bpm on the cool day. Together, the two residuals account for the whole 6 bpm gap in average heart rate, 151.1 versus 145.1.

Returning from injury (established runner): 10.9 to 15.6, Light. A runner recovering from a knee sprain follows a walk-run program. As their running time increases from 5 to 28 minutes over six weeks, their score gradually rises from 10.9 to 15.6. However, adding an 8-minute descent to that same 28-minute run jumps the score to 39.6 (**Moderate**), as grade and vertical oscillation label the downhill as **Hard** even though their heart rate stays calm.

Here TES shows how load builds: steady scores show a gradual progression, and the spike highlights biomechanical load that heart rate alone would miss.

5 Using TES

TES ships as part of Quiver, an open-source Swift package for statistics, linear algebra and machine learning, released under the Apache License 2.0. Quiver has no external dependencies, and the same code runs on iOS, macOS, watchOS, tvOS and visionOS, as well as on server-side Swift and Linux. It can be added to a new or existing project with Swift Package Manager:

```
dependencies: [
    .package(url: "https://github.com/waynewbishop/quiver", from: "1.6.0")
]
```

`TrueEffortScore` is constructed empty for a new runner, or decoded to resume a saved effort. Samples are recorded live, and the `finalize()` function closes the run:

```
import Quiver

var tes = TrueEffortScore()               // first run: no personal history yet

// For each sample, live or read back from HealthKit after the run:
tes.record(heartRate: 152, pace: 5.4, cadence: 174, grade: 1.2,
           verticalOscillation: 7.9, altitude: 100,
           at: sampleDate, hrTrust: 1.0)
// bpm, min/km, steps/min, percent, cm, meters; hrTrust is the
// app's confidence in the heart-rate reading, 0 to 1

tes.currentScore        // live score, before session adjustments
tes.currentEffort       // .easy, .tempo, .threshold or .hard
                        // (.recovery only for detected walking)
tes.currentResidual     // observed - expected HR, once a baseline exists

if let result = tes.finalize() {            // nil if the run is too short to keep
    result.adjusted                         // the headline score
    result.raw                              // before session adjustments
    result.averageHeartRate                 // average recorded HR, every run
    result.expectedHeartRate                // average expected HR, nil on a first run
    result.meanResidual                     // average residual in bpm, nil on a first run
    result.effortDistribution               // share of time in each category
    result.time(in: .easy)                  // seconds spent in Easy, 0 if never entered
}

if let baseline = tes.baseline {            // nil until a baseline has been learned
    baseline.equation()
    // "expected HR = 142.69 - 4.62·pace + 2.87·cadence + 4.85·grade
    //                - 3.95·verticalOscillation + 0.54·altitude"
}
```

6 Roadmap

The next stage of TES extends the model to more environments and more kinds of runners. Details include:

- **Surface complexity.** Exploring ways to detect the impact of roots, rocks, and loose footing. Currently, a technical trail scores similarly to smooth pavement; capturing surface complexity will let TES appropriately credit the extra effort required on challenging terrain.
- **Indoor elevation.** Investigating methods to incorporate treadmill incline data. For now, indoor runs default to a 0.5% grade, but capturing slope directly will let TES score indoor climbing efforts.
- **Environmental adaptation.** Refining how the model handles temperature and humidity on a runner's very first outings. Because high heat naturally elevates heart rate, the goal is to better isolate these environmental factors from the actual physical workload while a personal baseline is still being established.
- **Broadening profile support.** Expanding the model to automatically adapt to diverse physiological profiles. While the initial thresholds were calibrated to a single runner's absolute data, ongoing work aims to provide personalized scoring for runners with naturally lower maximum heart rates, such as older athletes or those on beta blockers.

6.1 Disclaimer

The True Effort Score is an experimental model for training analysis, research and education. It estimates training load from wearable sensor data. It is not a medical device, isn't intended to diagnose, treat, cure or prevent any condition, and doesn't provide medical advice. Scores, residuals and effort labels describe patterns in recorded data and may be inaccurate. Consult a qualified healthcare professional before starting or changing a training program, especially when returning from injury or managing a health condition.

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